Crude oil, alkanes and alkenes

Mark scheme and worked solutions for 30 questions: 10 multiple choice, 20 structured.

1 · Part 1

Paper 1 & 2 Introduction [1]

Answer: C

Worked solution

A homologous series is a family of compounds such as the alkanes or the alkenes. All members fit one general formula (for example CnH2n+2), each member differs from the next by CH_2, they have the same functional group, so they react in similar ways, and their physical properties such as boiling point show a gradual trend.

2 · Part 1

Paper 1 & 2 Introduction [1]

Answer: A

Worked solution

The empirical formula is the simplest whole-number ratio of atoms of each element. Divide both numbers in C_6H_{14} by their highest common factor, 2: 6 ÷ 2 = 3 and 14 ÷ 2 = 7. The empirical formula is C_3H_7.

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3 · Part 1 Paper 1 & 2 Introduction [5]

Mark scheme

(a) **[2]**

- (compounds with) the same molecular formula (1)
- but different structural formulae (1)

ALLOW different arrangements of atoms OR different structures for the second mark

REJECT same chemical formula but different properties alone, for the second mark

(b) **[1]**

- $CH_3CH(CH_3)CH_3$ (1)

ALLOW $(CH_3)_2CHCH_3$ OR $(CH_3)_3CH$ OR $CH(CH_3)_3$

(c) **[2]**

- C_4H_{10} (1)
- isomers have the same molecular formula OR it is an isomer of butane, which is C_4H_{10} (1)

ALLOW it has 4 carbon atoms and 10 hydrogen atoms (counted from its structural formula) for the second mark

Worked solution

(a) Isomers are compounds that have the same molecular formula but different structural formulae, so their atoms are arranged differently.

(b) 2-methylpropane has a chain of three carbon atoms (propane) with a methyl group, CH_3, on the second carbon: $CH_3CH(CH_3)CH_3$.

(c) Butane, $CH_3CH_2CH_2CH_3$, has 4 carbon atoms and 10 hydrogen atoms, so it is C_4H_{10}. Isomers always have the same molecular formula, so 2-methylpropane is also C_4H_{10}. Counting the atoms in $CH_3CH(CH_3)CH_3$ gives the same result.

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4 · Part 1 Paper 1 & 2 Introduction [6]

Mark scheme

(a) **[2]**

- reaction 1: combustion (1)
- reaction 2: addition (1)

(b) **[2]**

- an atom (or group of atoms) in a molecule is replaced by a different atom (or group) (1)
- in reaction 3, a hydrogen atom in methane is replaced by a chlorine atom (1)

ALLOW one hydrogen atom is swapped for a chlorine atom, forming CH_3Cl and HCl

IGNORE references to ultraviolet light

(c) **[2]**

- addition (1)
- two molecules join to form only one product OR hydrogen atoms are added to the C=C double bond (1)

ALLOW there is only one product, for the second mark

Worked solution

(a) Reaction 1 is ethane reacting with oxygen to form carbon dioxide and water, which is combustion. Reaction 2 is ethene and bromine joining to make a single product, which is addition.

(b) In a substitution reaction one atom in a molecule is replaced by a different atom. In reaction 3 one hydrogen atom of CH_4 is replaced by a chlorine atom to give CH_3Cl; the hydrogen atom that was removed ends up in HCl.

(c) Ethene and hydrogen join to form one product only, ethane, so this is an addition reaction. The hydrogen atoms add across the C=C double bond.

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5 · Part 1 Paper 1 & 2 Introduction [6]

Mark scheme

(a) **[4]**

- $CH_3CH_2CH(CH_3)CH_2CH_3$: 3-methylpentane (1)
- $CH_3CH(CH_3)CH(CH_3)CH_3$: 2,3-dimethylbutane (1)
- $CH_3CH_2CH{=}CHCH_3$: pent-2-ene (1)
- $CH_3CH_2CH_2CH_2Cl$: 1-chlorobutane (1)

ALLOW 2-pentene for pent-2-ene

IGNORE missing hyphens or commas if the numbers and name are otherwise correct

(b) **[2]**

- the chain should be numbered from the end nearest the methyl group (branch) OR the branch must be given the lowest possible number (1)
- 2-methylpentane (1)

Worked solution

(a) Find the longest carbon chain, then name and number the branches or functional group using the lowest possible numbers.

$CH_3CH_2CH(CH_3)CH_2CH_3$ has a chain of five carbons with a methyl group on carbon 3 (from either end): 3-methylpentane.

$CH_3CH(CH_3)CH(CH_3)CH_3$ has a chain of four carbons with methyl groups on carbons 2 and 3: 2,3-dimethylbutane.

$CH_3CH_2CH{=}CHCH_3$ has five carbons with the double bond starting at carbon 2 when numbered from the right: pent-2-ene.

$CH_3CH_2CH_2CH_2Cl$ has four carbons with chlorine on carbon 1: 1-chlorobutane.

(b) Priya numbered the chain from the wrong end. Numbering from the end nearest the branch puts the methyl group on carbon 2, so the correct name is 2-methylpentane.

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6 · Part 1 Paper 1 & 2 Introduction [6]

Mark scheme

(a) **[1]**

- an atom or group of atoms that determines the (characteristic) chemical reactions of a compound (1)

ALLOW the part of a molecule responsible for its reactions

(b) **[2]**

Any two (1 each):

- each formula differs from the next by CH_2 (1)
- there is a (steady) trend in boiling point OR boiling point increases as the molecules get bigger (1)
- all the formulae fit the same general formula, CnH2n (1)

(c) **[2]**

- C_5H_{10} (1)
- any value from 15 °C to 40 °C (1)

(d) **[1]**

- they have the same functional group OR they all contain a C=C double bond (1)

Worked solution

(a) A functional group is the atom or group of atoms that gives a compound its characteristic reactions. In alkenes it is the C=C double bond.

(b) Each formula has one more CH_2 than the one before (C_2H_4, C_3H_6, C_4H_8), they all fit CnH2n, and the boiling point rises steadily from −104 °C to −47 °C to −6 °C.

(c) Adding CH_2 to C_4H_8 gives C_5H_{10}. The boiling point rises by 57 °C and then by 41 °C, so the next rise is likely to be smaller again, about 25–35 °C. A sensible estimate is about 20–30 °C; the real value is about 30 °C.

(d) Chemical reactions depend on the functional group. All three compounds contain the C=C double bond, so they react in similar ways, for example with bromine.

7 · Part 1 Paper 1 & 2 Crude oil [1]

Answer: D

Worked solution

Bitumen has the largest molecules and highest boiling point of the fractions. It is a black, very viscous material that sets hard when cool, which makes it suitable for surfacing roads and roofs.

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8 · Part 1

Paper 1 & 2 | Crude oil | [1]

Answer: B

Worked solution

A hydrocarbon is a compound of carbon and hydrogen only. Crude oil contains many different hydrocarbons with different chain lengths and boiling points mixed together, which is why it can be separated into fractions.

9 · Part 1

Paper 1 & 2 | Crude oil | [1]

Answer: A

Worked solution

Carbon monoxide is colourless and has no smell. When it is breathed in, it combines with the blood in place of oxygen, so the blood can carry less oxygen around the body. This can cause drowsiness and, at high levels, death.

10 · Part 1

Paper 1 & 2 | Crude oil | [1]

Answer: C

Worked solution

Long-chain alkane vapour is passed over a hot catalyst of silica or alumina at 600–700 °C. The high temperature and the catalyst break the long molecules into shorter-chain alkanes and alkenes.

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11 · Part 1 Paper 1 & 2 Crude oil [7]

Mark scheme

(a) **[4]**

Indicative points, any four (1 each):

- crude oil is heated so that most of it vaporises (1)
- the vapour passes into the fractionating column (1)
- the column is hot at the bottom and cooler at the top OR there is a temperature gradient up the column (1)
- vapours rise up the column and condense when they reach a level that is cooler than their boiling point (1)
- fractions with different boiling points condense at different heights and are drawn off (1)
- fractions with lower boiling points (smaller molecules) are collected near the top OR higher boiling points near the bottom (1)

(b) **[2]**

- bitumen has the highest boiling point OR has the largest molecules (1)
- so it does not vaporise OR it stays as a liquid (and flows to the bottom of the column) (1)

(c) **[1]**

- it is a mixture of different hydrocarbons, which have different boiling points (1)

ALLOW it contains more than one compound

Worked solution

(a) Crude oil is heated until most of it turns to vapour. The vapour enters a tall fractionating column, which is hottest at the bottom and coolest at the top. As the vapours rise they cool. Each hydrocarbon condenses when it reaches a level where the temperature is below its boiling point. Hydrocarbons with similar boiling points condense together and are drawn off as one fraction. Small molecules with low boiling points rise highest; large molecules with high boiling points condense lower down.

(b) Bitumen has the largest molecules and the highest boiling point, so it does not boil even at the bottom of the column. It stays as a liquid and is collected there.

(c) A fraction is not a pure substance. Gasoline contains many hydrocarbons with similar, but not identical, boiling points, so it boils over a range of temperatures.

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12 · Part 1 Paper 1 & 2 Crude oil [5]

Mark scheme

(a) **[3]**

- fuel for aircraft: kerosene (1)
- bottled gas for heating and cooking: refinery gases (1)
- fuel for large ships and some power stations: fuel oil (1)

ALLOW paraffin for kerosene; LPG for refinery gases

(b) **[1]**

- fuel for lorries OR buses OR trains OR cars with diesel engines (1)

ALLOW fuel for diesel engines

(c) **[1]**

- fuel for cars (1)

ALLOW petrol for cars

Worked solution

(a) Kerosene is used as aircraft fuel. Refinery gases (such as propane and butane) are bottled and used for heating and cooking. Fuel oil is burned in large ships and some power stations.

(b) Diesel is used as fuel in diesel engines, such as those in lorries, buses, trains and some cars.

(c) Gasoline (petrol) is the fuel for cars with petrol engines.

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13 · Part 1

Paper 1 & 2 Crude oil [6]

Mark scheme

(a) **[1]**

- a substance that releases heat energy when it is burned (1)

ALLOW a substance that is burned to release energy

(b) **[2]**

- 3 (CO_2) and 4 (H_2O) (1)
- 5 (O_2) (1)

ALLOW the fully correct equation $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$ for both marks

(c) **[2]**

- carbon monoxide (1)
- carbon OR soot (1)

REJECT carbon dioxide; water

(d) **[1]**

- a yellow (or orange) flame OR a smoky flame OR black soot forms on the stove (or pan) (1)

REJECT a blue flame

Worked solution

(a) A fuel is a substance that releases heat energy when it burns.

(b) Balance carbon first: 3 C atoms give $3CO_2$. Then hydrogen: 8 H atoms give $4H_2O$. Count the oxygen atoms on the right: $(3 \times 2) + 4 = 10$, so $5O_2$ is needed. $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$.

(c) With too little oxygen, not all the carbon is fully oxidised. Carbon monoxide, CO, and carbon (soot), C, can form. Neither forms when there is plenty of oxygen.

(d) A yellow, smoky flame, or black soot on the bottom of a pan, shows that unburned carbon is being produced.

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14 · Part 1 Paper 1 & 2 Crude oil [5]

Mark scheme

(a) **[1]**

- 2 ($N_2 + O_2 \rightarrow 2NO$) (1)

(b) **[2]**

- both come from the air (1)
- the temperature in the engine is high enough for them to react OR the air around us is not hot enough (1)

REJECT nitrogen comes from the fuel

(c) **[2]**

- oxides of nitrogen dissolve in OR react with water (and oxygen) in the atmosphere (clouds or rain) (1)
- forming nitric acid OR an acid (which falls as acid rain) (1)

ALLOW nitrous acid

Worked solution

(a) There are 2 nitrogen atoms and 2 oxygen atoms on the left, so 2 molecules of NO form: $N_2 + O_2 \rightarrow 2NO$.

(b) Air is mostly nitrogen, with oxygen. Nitrogen is very unreactive, but the temperature inside a car engine is high enough for nitrogen and oxygen to react. The air around us is far too cool for this.

(c) The oxides of nitrogen leave in the exhaust gases and rise into the atmosphere. They dissolve in water droplets in clouds and react to form nitric acid, which makes the rain acidic.

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15 · Part 1 Paper 1 & 2 Crude oil [7]

Mark scheme

(a) **[2]**

- the fuel contains sulfur (or sulfur compounds) as an impurity (1)
- the sulfur reacts with oxygen (from the air) when the fuel burns (1)

REJECT hydrocarbons contain sulfur

(b) **[1]**

- $S + O_2 \rightarrow SO_2$ (1)

IGNORE state symbols

(c) **[2]**

- fuel Y (1)
- it contains less sulfur, so less sulfur dioxide is produced (when it burns) (1)

REJECT a reason that does not refer to less sulfur or less sulfur dioxide

(d) **[2]**

- Mr of SO_2 = 64 OR mass ratio S:SO_2 = 32:64 OR 2000 g ÷ 32 = 62.5 mol (of S, so 62.5 mol of SO_2) (1)
- 4.0 kg (1)

ALLOW 4 kg OR 4000 g

ECF from an incorrect Mr of SO_2

Worked solution

(a) Some diesel fuels contain sulfur, or compounds of sulfur, as impurities. When the fuel burns, the sulfur reacts with oxygen from the air to form sulfur dioxide.

(b) $S + O_2 \rightarrow SO_2$. One sulfur atom and one oxygen molecule form one molecule of sulfur dioxide.

(c) Fuel Y contains 0.010 kg of sulfur per 1000 kg, much less than fuel X (2.0 kg). Less sulfur means less sulfur dioxide, and sulfur dioxide dissolves in rainwater to form acid rain, so fuel Y causes less acid rain.

(d) Ar of S = 32 and Ar of O = 16, so Mr of SO_2 = 32 + (2 × 16) = 64. One mole of S (32 g) forms one mole of SO_2 (64 g), so the mass doubles. 2.0 kg of sulfur forms 2.0 × 64 ÷ 32 = 4.0 kg of sulfur dioxide. Check in grams: 2000 g ÷ 32 = 62.5 mol; 62.5 × 64 = 4000 g = 4.0 kg.

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16 · Part 1 Paper 1 & 2 Crude oil [6]

Mark scheme

(a) **[1]**

- cracking needs a high temperature OR to provide energy to break the long-chain alkane molecules (1)

ALLOW so that the catalyst is hot enough to crack the paraffin vapour

(b) **[1]**

- it is (mainly) air (displaced from the apparatus) (1)

(c) **[2]**

- (otherwise) water is sucked back (up the delivery tube) into the boiling tube (1)
- (cold water hitting) the hot glass could crack it OR the tube could shatter (1)

ALLOW suck-back for the first mark

(d) **[2]**

- C_4H_8 (1)
- it fits the general formula CnH2n OR it has two fewer hydrogen atoms than an alkane with 4 carbons OR it must contain a C=C double bond (1)

ALLOW C_6H_{14} fits CnH2n+2 so it is the alkane, for the second mark

Worked solution

(a) Cracking only happens at a high temperature. Heating the aluminium oxide strongly means the paraffin vapour is broken down into smaller molecules as it passes over the hot catalyst.

(b) The first gas pushed out of the apparatus is the air that was already inside it, so it is not the cracking product.

(c) If the heating stops while the delivery tube is still in the water, the gas inside cools and contracts. Water is then sucked back into the hot boiling tube, and the sudden cooling can crack the glass.

(d) C_6H_{14} fits CnH2n+2 ($2 \times 6 + 2 = 14$), so it is an alkane. C_4H_8 fits CnH2n ($2 \times 4 = 8$), so it is the alkene; it has a C=C double bond.

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17 · Part 1 Paper 1 & 2 Crude oil [7]

Mark scheme

(a) **[1]**

- gasoline (1)

(b) **[4]**

Indicative points, any four (1 each); maximum 3 if no data from the table is used:

- the supply of fuel oil and bitumen (55%) is greater than the demand (38%) (1)
- the demand for gasoline (25%) is greater than the supply (10%) OR demand for diesel (23%) is greater than supply (18%) (1)
- cracking breaks long-chain alkanes (from fuel oil and bitumen) into shorter-chain alkanes (1)
- the shorter-chain alkanes can be used as gasoline OR this increases the supply of gasoline to meet demand (1)
- this uses up the surplus of the long-chain fractions (1)
- cracking also produces alkenes, which are useful (for example to make polymers or other chemicals) (1)

(c) **[2]**

- $20 = 8 + 2x$ OR $42 = 18 + 4x$ OR $(20 - 8) \div 2$ (1)
- 6 (1)

ALLOW a correct balanced equation $C_{20}H_{42} \rightarrow C_8H_{18} + 6C_2H_4$ for both marks

Worked solution

(a) Demand minus supply: refinery gases $6 - 4 = 2$, gasoline $25 - 10 = 15$, diesel $23 - 18 = 5$. Gasoline has the largest shortage.

(b) The crude oil supplies far more of the long-chain fractions than people want: fuel oil and bitumen make up 55% of the supply but only 38% of the demand. Short-chain fractions such as gasoline are in short supply: 10% supply but 25% demand. Cracking breaks the surplus long-chain alkanes into shorter-chain alkanes that can be used as gasoline, so supply matches demand better. Cracking also makes alkenes, which are used to make polymers and other chemicals.

(c) Carbon atoms: $20 = 8 + 2x$, so $2x = 12$ and $x = 6$. Check with hydrogen atoms: $18 + (6 \times 4) = 42$. $C_{20}H_{42} \rightarrow C_8H_{18} + 6C_2H_4$.

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18 · Part 1 Paper 1 & 2 Crude oil [6]

Mark scheme

(a) **[1]**

- water (1)

(b) **[2]**

- limewater turns milky OR cloudy (1)
- carbon dioxide (1)

(c) **[2]**

- water is formed, so candle wax contains hydrogen (1)
- carbon dioxide is formed, so candle wax contains carbon (1)

(d) **[1]**

- the wax might also contain oxygen (as the oxygen in the products could come from the wax as well as from the air) OR the test does not show that no other elements are present (1)

Worked solution

(a) Water turns white anhydrous copper(II) sulfate blue, so the liquid is water.

(b) Carbon dioxide turns limewater milky (cloudy).

(c) The hydrogen in the water and the carbon in the carbon dioxide must have come from the wax, because air does not supply carbon or hydrogen. So candle wax contains carbon and hydrogen.

(d) A hydrocarbon contains carbon and hydrogen only. The oxygen in the products could have come from the air or from the wax, so the results cannot rule out oxygen (or other elements) in the wax.

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19 · Part 1 Paper 1 & 2 Crude oil [6]

Mark scheme

(a) **[1]**

- the more carbon atoms in the molecules, the higher the boiling range (1)

(b) **[2]**

- larger molecules have stronger forces between the molecules (1)
- so more (heat) energy is needed to overcome these forces (1)

ALLOW intermolecular forces

REJECT breaking covalent bonds OR stronger bonds within the molecules

(c) **[2]**

- colour: becomes darker (1)
- viscosity: increases OR becomes thicker (1)

(d) **[1]**

- kerosene (1)

Worked solution

(a) Refinery gases (1–4 carbon atoms) boil below 30 °C, while fuel oil (20–40 carbon atoms) boils above 340 °C. More carbon atoms means a higher boiling range.

(b) Boiling separates whole molecules from each other; it does not break the covalent bonds inside them. Larger molecules have stronger forces between them, so more energy is needed to overcome these forces, and the boiling point is higher.

(c) Going from gasoline to fuel oil, the fractions become darker in colour and more viscous (thicker, flowing less easily).

(d) 13 carbon atoms lies in the range 10–16 (kerosene) but not 5–10 or 14–20, so $C_{13}H_{28}$ is in the kerosene fraction.

20 · Part 1 Paper 1 & 2 Alkanes [1]

Answer: D

Worked solution

Use CnH2n+2 with n = 9: H = 2 × 9 + 2 = 20. The molecular formula is C_9H_{20}.

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21 · Part 1 Paper 1 & 2 Alkanes [1]

Answer: B

Worked solution

A saturated hydrocarbon contains only carbon and hydrogen, and has only single covalent bonds between carbon atoms. Look for a formula with no "=" sign and no atoms other than C and H: only $CH_3CH_2CH_3$ fits.

22 · Part 1 Paper 1 & 2 Alkanes [4]

Mark scheme

(a) **[1]**

- pentane (1)

(b) **[2]**

- $CH_3CH(CH_3)CH_2CH_3$ (1)
- $C(CH_3)_4$ (1)

ALLOW $(CH_3)_2CHCH_2CH_3$ OR $CH_3CH_2CH(CH_3)CH_3$ for the first; $CH_3C(CH_3)_2CH_3$ OR $(CH_3)_4C$ for the second

ALLOW the two formulae in either order

(c) **[1]**

- 16 (1)

Worked solution

(a) The unbranched isomer has all five carbon atoms in one chain: $CH_3CH_2CH_2CH_2CH_3$, pentane.

(b) Moving one carbon to a branch gives a four-carbon chain with a methyl group on carbon 2: $CH_3CH(CH_3)CH_2CH_3$ (2-methylbutane). Moving two carbons gives a three-carbon chain with two methyl groups on the middle carbon: $C(CH_3)_4$ (dimethylpropane).

(c) Each molecule has 12 C–H bonds (one for each hydrogen atom) and 4 C–C bonds (five carbon atoms joined without rings need four links). 12 + 4 = 16 covalent bonds. This is the same for all three isomers.

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23 · Part 1 Paper 1 & 2 Alkanes [5]

Mark scheme

(a) **[1]**

- C_3H_7Br (1)

(b) **[2]**

- $CH_3CH_2CH_2Br$ (1)
- $CH_3CHBrCH_3$ (1)

ALLOW $CH_2BrCH_2CH_3$ for the first; $(CH_3)_2CHBr$ for the second

ALLOW the two formulae in either order

(c) **[2]**

- 1-bromopropane (1)
- 2-bromopropane (1)

REJECT a name that does not match the structure given in (b)

ECF a correct name for an incorrect structure from (b), only if that structure has one bromine atom on a three-carbon chain

Worked solution

(a) One hydrogen atom in C_3H_8 is replaced by one bromine atom: $C_3H_8 + Br_2 \rightarrow C_3H_7Br + HBr$. The organic product is C_3H_7Br.

(b) The bromine atom can replace a hydrogen on an end carbon, giving $CH_3CH_2CH_2Br$, or a hydrogen on the middle carbon, giving $CH_3CHBrCH_3$. Replacing a hydrogen on the other end carbon gives the same compound as $CH_3CH_2CH_2Br$.

(c) Number the chain so the bromine has the lowest number: $CH_3CH_2CH_2Br$ is 1-bromopropane and $CH_3CHBrCH_3$ is 2-bromopropane.

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24 · Part 1 Paper 1 & 2 Alkanes [5]

Mark scheme

(a) **[1]**

- $C_{12}H_{26}$ (1)

(b) **[1]**

- 170 (1)

ECF from an incorrect formula in (a)

(c) **[1]**

- C_6H_{13} (1)

ECF from an incorrect formula in (a)

(d) **[2]**

- alkanes have the general formula CnH2n+2 (1)
- so the number of hydrogen atoms is always more than twice the number of carbon atoms OR the ratio of C to H can never simplify to 1:2 (1)

ALLOW CH_2 is the empirical formula of compounds with the general formula CnH2n (alkenes), for the second mark

Worked solution

(a) Alkanes are CnH2n+2. 2n + 2 = 26, so 2n = 24 and n = 12. The formula is $C_{12}H_{26}$.

(b) Mr = (12 × 12) + (26 × 1) = 144 + 26 = 170.

(c) 12 and 26 can both be divided by 2, giving C_6H_{13}. 6 and 13 have no common factor, so this is the simplest ratio.

(d) In CnH2n+2 there are always 2 more hydrogen atoms than twice the number of carbon atoms, so the ratio C:H is never exactly 1:2. The empirical formula CH_2 means a ratio of exactly 1:2, which fits CnH2n (the alkenes), not the alkanes.

25 · Part 1 Paper 1 & 2 Alkenes [1]

Answer: C

Worked solution

For each formula, double the number of carbon atoms and compare with the number of hydrogen atoms. Only C_7H_{14} has exactly twice as many hydrogen atoms as carbon atoms, so only C_7H_{14} could be an alkene.

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26 · Part 1 Paper 1 & 2 Alkenes [1]

Answer: A

Worked solution

In propene, $CH_2=CHCH_3$, the double bond is between the first and second carbon atoms. In an addition reaction the double bond opens and one bromine atom bonds to each of these two carbons: $CH_2BrCHBrCH_3$.

27 · Part 1 Paper 1 & 2 Alkenes [5]

Mark scheme

(a) **[2]**

- $CH_2=CHCH_2CH_3$ (1)
- $CH_3CH=CHCH_3$ (1)

ALLOW $CH_3CH_2CH=CH_2$ for the first

ALLOW the two formulae in either order

(b) **[2]**

- but-1-ene (1)
- but-2-ene (1)

REJECT a name that does not match the structure given in (a)

ALLOW 1-butene; 2-butene

REJECT butene without a number

(c) **[1]**

- $CH_2=C(CH_3)CH_3$ (1)

ALLOW $CH_2=C(CH_3)_2$ OR $(CH_3)_2C=CH_2$

Worked solution

(a) With four carbons in a straight chain, the double bond can be at the end of the chain, $CH_2=CHCH_2CH_3$, or in the middle, $CH_3CH=CHCH_3$. A double bond at the other end, $CH_3CH_2CH=CH_2$, is the same molecule as the first one.

(b) Number the chain from the end nearest the double bond. $CH_2=CHCH_2CH_3$ is but-1-ene; $CH_3CH=CHCH_3$ is but-2-ene.

(c) A branched chain has three carbons in the main chain and a CH_3 branch. Putting the double bond so that every carbon still has four bonds gives $CH_2=C(CH_3)CH_3$ (methylpropene). It is also C_4H_8.

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28 · Part 1 Paper 1 & 2 Alkenes [4]

Mark scheme

(a) **[3]**

- add bromine water to each compound and shake (1)
- hex-1-ene: the bromine water changes from orange (or yellow or brown) to colourless OR is decolourised (1)
- hexane: the bromine water stays orange OR no colour change (1)

REJECT clear for colourless

REJECT hexane turns colourless

(b) **[1]**

- addition (1)

Worked solution

(a) Add a few drops of bromine water to a sample of each liquid and shake. With hex-1-ene the orange colour disappears and the mixture becomes colourless, because bromine reacts with the C=C double bond. With hexane there is no double bond, so the bromine does not react (at room temperature, without ultraviolet light) and the mixture stays orange.

(b) The bromine atoms add across the C=C double bond of hex-1-ene to form a single product, so this is an addition reaction.

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29 · Part 1 Paper 1 & 2 Alkenes [6]

Mark scheme

(a) **[2]**

- ethene contains a carbon–carbon double bond OR C=C (1)
- so more atoms can be added to it OR it does not contain the maximum number of hydrogen atoms possible OR it contains only carbon and hydrogen (hydrocarbon) (1)

(b) **[1]**

- 1,2-dibromoethane (1)

REJECT dibromoethane without numbers; 1,1-dibromoethane

(c) **[3]**

- Mr of C_2H_4 = 28 AND amount = 2.8 ÷ 28 = 0.10 mol (1)
- Mr of Br_2 = 160 (1)
- mass of bromine = 0.10 × 160 = 16 g (1)

ALLOW mass ratio method, 2.8 × 160 ÷ 28 = 16 g

ECF from an incorrect amount of ethene or an incorrect Mr

Worked solution

(a) Ethene contains only carbon and hydrogen, so it is a hydrocarbon. It has a C=C double bond, so it is unsaturated: it can take part in addition reactions in which more atoms join on to the carbon atoms.

(b) One bromine atom adds to each carbon atom, giving CH_2BrCH_2Br, which is 1,2-dibromoethane.

(c) Mr of C_2H_4 = (2 × 12) + (4 × 1) = 28. Amount of ethene = 2.8 ÷ 28 = 0.10 mol. The equation shows 1 mol of ethene reacts with 1 mol of Br_2, so 0.10 mol of Br_2 reacts. Mr of Br_2 = 2 × 80 = 160. Mass = 0.10 × 160 = 16 g. Check: 2.8 g is one tenth of 28 g, so the bromine is one tenth of 160 g, which is 16 g.

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30 · Part 1 Paper 1 & 2 Alkenes [6]

Mark scheme

(a) **[2]**

- alkene general formula CnH2n so 12n + 2n = 84 OR 14n = 84 (1)
- n = 6, so C_6H_{12} (1)

ALLOW a correct trial of C_6H_{12} = (6 × 12) + (12 × 1) = 84 for both marks

(b) **[2]**

- alkane general formula CnH2n+2 so 12n + 2n + 2 = 100 OR 14n + 2 = 100 (1)
- n = 7, so C_7H_{16} (1)

ALLOW a correct trial of C_7H_{16} = (7 × 12) + (16 × 1) = 100 for both marks

(c) **[2]**

- CH_2=$CHCH_2CH_2CH_3$ (1)
- pent-1-ene (1)

ALLOW $CH_3CH_2CH_2CH$=CH_2; 1-pentene

ALLOW the name mark for pent-1-ene even if the formula is incorrect

Worked solution

(a) Alkenes are CnH2n. Each carbon contributes 12 and each hydrogen 1, so Mr = 12n + 2n = 14n. 14n = 84 gives n = 6, so J is C_6H_{12}. Check: 72 + 12 = 84.

(b) Alkanes are CnH2n+2, so Mr = 14n + 2. 14n + 2 = 100 gives 14n = 98 and n = 7, so K is C_7H_{16}. Check: 84 + 16 = 100.

(c) For L, 14n = 70 gives n = 5, so L is C_5H_{10}. With an unbranched chain and the double bond between carbons 1 and 2, the structural formula is CH_2=$CHCH_2CH_2CH_3$ and the name is pent-1-ene.

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